CHRONO-1 — Preregistration

Series: CHRONO (temporal nonclassicality)
Experiment ID: CHRONO-1
Owner: Remnant Fieldworks Inc. (Derek Adam Hone)
Status at lock: design + harness complete, validated in simulation, hardware job pending
Governing framework: RF Experiment–Pilot–IP Covenant (preregistration-first; honest outcomes; claims narrower than evidence; file-first / publish-after for new matter)

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1. One-sentence statement

We measure the three-time **Leggett–Garg inequality** (temporal Bell-type inequality) on a single superconducting qubit and **cryptographically bind the measured K3 statistic into an authorization ProofRecord** as a temporal nonclassicality witness — a certificate that the authorized value had **no definite pre-measurement (pre-execution) history** under the stated assumption.

2. Why this is new (distinct from all prior RF series)

Series	Phenomenon tested	Resource witnessed
ARK	authorization **boundary** (context replay)	exact-match policy
WITNESS	provenance of a superposition-sourced nonce	assumed unpredictability
BELLWETHER	**spatial** nonclassicality (CHSH Bell, S>2)	Bell violation bound into nonce
CHRONO	**temporal** nonclassicality (Leggett–Garg K3>1)	time-ordered nonclassicality bound into nonce

No prior work (RF or, to our knowledge, anyone) binds a **Leggett–Garg temporal-correlation witness** into an authorization/ProofRecord construction. BELLWETHER witnessed nonclassicality across space; CHRONO witnesses it across time — the physical statement most directly aligned with RF doctrine: **Verification Before Execution** (the value did not exist definitely before it was verified).

3. Physics and design

Single qubit, dichotomic observable **Q = Z** (outcome +1 for $|0\rangle$, −1 for $|1\rangle$). Between measurement times the qubit precesses under **R_y(θ)** with **θ = π/3 (60°)**.

Three two-time correlators, each measured in a **separate circuit** using **mid-circuit measurement**:

```
C12 : measure(t1) -> R_y(θ) -> measure(t2)
C23 : R_y(θ) -> measure(t2) -> R_y(θ) -> measure(t3)
C13 : measure(t1) -> R_y(2θ) -> measure(t3)
```

Each circuit writes two bits into one 2-bit classical register `c` (`c[0]` = earlier, `c[1]` = later). Correlator = shot-average of the product of the two signed outcomes.

Leggett–Garg K3: `K3 = C12 + C23 − C13`

- **Macrorealist bound (H1):** `K3 ≤ 1`
- **Quantum (Lüders) bound:** `K3 = 1.5` (maximal at θ = π/3)
- Ideal predictions: `C12 = C23 = cos60° = +0.5`, `C13 = cos120° = −0.5`, so `K3 = 1.5`.

Simulation validation (pre-lock, noiseless AerSimulator, 200k shots/correlator): `C12 = 0.502, C23 = 0.503, C13 = −0.504, K3 = 1.508` — consistent with the ideal 1.5; macrorealist bound violated at ~152σ. The counts-extraction path is confirmed identical to the hardware path.

4. Execution plan

- **Backend:** IBM open-plan hardware (prefer `ibm_fez`, else min pending-jobs).
- **Shots:** 2000 per correlator × 3 = **6000 shots total** (~a few seconds of QPU time; sized to the remaining open-plan budget).
- **Mode:** Batch (open plan forbids Session). **Channel:** `ibm_quantum_platform`.
- **Circuits transpiled** to backend ISA (optimization_level=1) before submission.

5. Outcome criteria (fixed in advance — honest, symmetric)

Verdict	Condition
VIOLATION (temporal nonclassicality, under assumptions)	`K3 > 1` and `K3 − 1 > 3σ`
INCONCLUSIVE / HOLD	`K3 > 1` but not 3σ-significant
NO VIOLATION	`K3 ≤ 1`

The measured hardware K3 will be reported **exactly as observed**, whichever bin it lands in. No re-runs to chase a preferred number.

6. Honesty boundary (non-negotiable — carried into every artifact)

- The Leggett–Garg inequality assumes **non-invasive measurability (NIM)**: that a macrorealist system can be read without disturbance. A projective qubit measurement **is** invasive, so a determined macrorealist can attribute any `K3 > 1` to measurement disturbance — the **"clumsiness loophole."**
- This is therefore a **Leggett–Garg violation UNDER STATED ASSUMPTIONS**, **not** a loophole-free refutation of macrorealism, and **not** device-independent certified randomness.

- **Single system**, single device: no device-independence is claimed.
- **Only defensible claim:** “observed temporal correlations violating the macrorealist bound K3 ≤ 1 on hardware, under the non-invasive-measurability assumption, cryptographically bound into the authorization record.”

6a. Disclosed pre-execution harness facts

- Local simulator (`StatevectorSampler`) cannot execute mid-circuit measurements; validation used `AerSimulator` . The **hardware** path (SamplerV2, Batch) executes mid-circuit measurement natively. This does not alter any circuit or claim.
- Secrets key renamed `witness1_ibmq` → `ibm quantum` ; the token loader tries both (FIX-3). No physics impact.

7. ProofRecord construction (`chrono-proofrecord-1.0`)

```
raw_counts_hash  = H( canonical(counts_by_correlator) )
calibration_hash = H( canonical(calibration_snapshot) )
witness_hash     = H( canonical(temporal_witness) )     # contains K3, correlators,
bounds
chrono_nonce     = H( lp(raw_counts_hash) || lp(job_id) || lp(calibration_hash) ||
lp(witness_hash) )
record_hash      = H( canonical(all fields except record_hash) )
```

Tampering with the measured statistic breaks `chrono_nonce` ; any covered-field edit breaks `record_hash` . Verified by 9/9 pre-lock mock tests (tamper-evidence over both counts and witness) and by the independent `verify.py` .

8. Commercial purpose (Covenant requirement)

CHRONO extends the ExecutionProof patent family with a **temporal** entropy/authorization primitive: a ProofRecord whose seed is certified to have possessed no definite value prior to its verification event. Candidate RF-100 use: **anti-replay / freshness certificates in time-ordered authorization** (a token that provably could not have been precomputed). Publication is gated pending the file-first IP determination (§9).

9. IP gate

CHRONO-1 is **almost certainly new patentable matter**. Per the Covenant it is held **local-only** (no GitHub / Zenodo / LinkedIn / RF-100 disclosure) until Derek makes the **file-first** determination. Execution is **not** disclosure; running the job and fetching results is permitted.

To God be the glory. Proof Before Power. Verification Before Execution.